A Neuro-Symbolic Expert System that Learns the Rules: Fusing a Differentiable Rule Layer with Neural Tabular Models for Interpretable, Calibrated, and Robust Decision Support

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Abstract

Rule-based expert systems are interpretable, auditable, and data-efficient, but they cannot learn from data and degrade when the encoded knowledge is approximate. Deep tabular models are accurate and flexible, but they are opaque, hard to calibrate, and brittle under distribution shift. We study how far the two paradigms can be reconciled in a single, deployable model and introduce NEXS (*Neuro-symbolic EXpert System*), which couples a differentiable rule layer—whose per-rule confidences are initialised and regularised towards an expert's certainty factors—with a neural tabular head, fused through an additively-gated logit. Unlike semantic-loss approaches that use knowledge only as a soft consistency penalty on the output, NEXS keeps the symbolic computation in the forward pass, so the learned logic remains explicit and recoverable. We build two controlled benchmarks—clinical triage and industrial fault diagnosis—in which the data are generated by a *known* Horn rule base whose intermediate concepts are hidden, enabling the simultaneous measurement of task accuracy and *rule recovery*. Across three seeds, NEXS recovers the most ground-truth rules of any learned model (rule-set F1 up to 0.33 versus 0.11–0.22 for an MLP and 0 for logistic regression) and attains the highest logic fidelity to the ground-truth oracle among learned models (0.86–0.91, approaching the expert system's 0.95–0.97). It matches the MLP on accuracy and is the best learned model in low-data regimes—up to +7 F1 points over the MLP at 300 training samples— while running at $\sim$0.6 μs per sample, about 110$\times$ faster than a random forest. We also find that a classical certainty-factor expert system, despite being the most logic-faithful and shift-robust predictor, is the *worst* calibrated (ECE $\sim$0.09), because its certainty factors are heuristics rather than probabilities; isotonic calibration makes all learned models well calibrated. Finally, we repeat the study on three small public datasets—UCI Cleveland heart disease, Pima diabetes, and tic-tac-toe—with hand-written knowledge bases. There the expert system is exact only when its knowledge is exact (tic-tac-toe, where it is the eight winning lines) and is the *weakest* learned-comparable method when the knowledge is approximate; with class weighting NEXS is the most accurate model on heart (0.836) and the best-calibrated on tic-tac-toe (ECE 0.002), while retaining the most expert logic of any learned model. Finally, we isolate *when* expert rules help: in a 675-run factorial study crossing knowledge-base corruption against training data, the expert prior is a monotone, positive contribution while the knowledge is intact—largest when data is scarce, worth more than additional data when the rules are correct—and turns negative once the knowledge is removed; a 20-seed paired analysis with Holm correction shows NEXS's advantage over the MLP and semantic loss is statistically significant ($p < 0.01$) on the dataset

with genuinely correct rules, while on the approximate-KB datasets the two are statistically indistinguishable. We release code, data, and a fully reproducible experiment suite.

1 Introduction

Expert systems encode human knowledge as explicit rules and dominated applied AI in the 1970s–80s, from MYCIN's certainty-factor calculus [20] to large diagnostic shells. Their strengths are well understood: decisions are traceable, they work with little or no data, and they degrade gracefully when rules are approximate. Their weaknesses are equally well understood: they cannot learn from data, knowledge acquisition is a bottleneck, and a wrong rule is a permanent error. The deep-learning era inverted the trade-off. Gradient-boosted trees and multilayer perceptrons now dominate tabular decision tasks, but they are opaque, miscalibrated [9], and brittle to distribution shift.

The third wave of neuro-symbolic AI [8, 12] argues that these paradigms should be combined rather than opposed. A large body of work does so by *compiling* knowledge into a differentiable loss: semantic loss [22], logic-rule distillation [10], and DL2 [5] all penalise outputs that violate the knowledge base during training. Others embed symbolic reasoning directly in the computation graph, as in DeepProbLog [15] and logic tensor networks [1], or inject rules into the architecture, as in the KBANN lineage [21, 6]. Yet a practical question remains open: *for the tabular decision-support problems that industry actually deploys, which coupling is worth its complexity, and does the resulting model still behave like the expert system it replaces?*

Answering this requires measuring more than accuracy. A model that replaces an expert system must (i) be at least as accurate, (ii) be calibrated enough to threshold safely, (iii) remain robust when the input distribution moves, and (iv) *learn logic that an expert can inspect and trust.* Most neuro-symbolic evaluations report (i) and sometimes (ii) or (iii); almost none report (iv), because doing so needs ground-truth rules, which real datasets rarely provide.

Contributions.

- We design two reproducible benchmarks—*clinical triage* and *industrial fault diagnosis*—in which data are generated by a known Horn rule base with *hidden* intermediate concepts, giving a ground-truth oracle for both accuracy and logic (Section 3).
- We introduce NEXS, a neuro-symbolic expert system that keeps a differentiable rule layer in the forward pass and anchors each rule's confidence to an expert certainty factor via a prior regulariser (Section 4.4).
- We run a seven-experiment study (Section 5) comparing a certainty-factor expert system, four classical learners, an MLP, a semantic-loss model, and NEXS, measuring accuracy, calibration, rule recovery, logic fidelity, data efficiency, shift robustness, and latency.
- We show that NEXS recovers more ground-truth rules and is more logic-faithful than any other learned model, matches the MLP on accuracy, and is strongest in low-data regimes, while the discrete expert system is the most logic-faithful and shift-robust but the *worst* calibrated predictor (Section 6).

2 Related Work

Neural-symbolic integration. The field is surveyed in [2, 7, 13, 16]. Approaches differ in *where* knowledge enters. Loss-based methods add a knowledge penalty to the objective: semantic loss [22] measures the probability mass of worlds satisfying a constraint; rule distillation [10] projects predictions onto a rule-regularised space; DL2 [5] trains with logical constraints as inequalities. Architecture-based methods build rules into the network: KBANN [21] initialises a network from propositional rules, and logic tensor networks [1] ground first-order logic in a real-valued tensor calculus. Probabilistic methods such as DeepProbLog [15] couple neural predicates with a logic program and perform inference in a probabilistic logic. NEXS sits between the loss-based and architecture-based families: like KBANN it uses the knowledge base to structure and initialise the computation, but it does so through a *differentiable, continuously-relaxed* rule layer whose confidences stay tied to the expert's certainty factors.

Rule extraction and interpretability. Post-hoc explainers such as LIME [18] and SHAP approximate a black box locally; recent work extends these to multivariate time series [14]. Extracting rules by distillation, as we do to evaluate logic recovery, is an established evaluation device but is rarely compared against ground-truth rules. Our rule-recovery metric closes that gap.

Hybrid AI and expert systems. Surveys of hybrid AI catalogue how rules and models are coupled in deployed systems [19], and applied reviews emphasise the need for interpretability in high-stakes domains [4]. We contribute a quantitative, multi-criteria comparison on controlled data.

3 Problem Formulation and Benchmarks

We study binary decision support where the label is the disjunction of a small number of conjunctive rules over *observed features* and *latent intermediate concepts*. Formally, a domain is a tuple $\langle \mathbf{f}, \mathbf{c}, \mathcal{R} \rangle$ where $\mathbf{f}$ are observed features, $\mathbf{c}$ are hidden concepts, and $\mathcal{R}$ is a Horn rule base whose heads are concepts or the label. Data are generated by first sampling features from a two-regime mixture (a shared latent propensity drives a "healthy" and an "at-risk" regime per feature), then realising concepts from the concept rules, then the label from the label rules, with small stochastic realisation and label noise. Concepts are never revealed to any model.

Domains. *Clinical triage* (8 features: age, systolic BP, cholesterol, max heart rate, ST depression, BMI, smoking, diabetes; concepts: hypertension, dyslipidemia) predicts high cardiac risk, in the spirit of the UCI heart-disease task [11]. *Industrial fault diagnosis* (6 sensors: temperature, pressure, vibration, flow, current, RPM; concepts: overheat, bearing wear) predicts an equipment fault, following predictive-maintenance benchmarks [17]. Each domain has five ground-truth rules, three of which produce the label. An *expert* rule base with human-plausible (rounded) thresholds and certainty factors is given to the symbolic and neuro-symbolic methods; it deliberately differs from the ground truth, so we can measure tolerance to imperfect knowledge.

4 Methods

4.1 Symbolic expert system (ESS)

We implement forward chaining in two layers. Concept confidences are computed from concept rules, then the label from concept and feature rules. A literal's membership is crisp (a step function) or soft (a tanh ramp), and contributions to the same head are combined with the MYCIN certainty-factor calculus [20]: $\mathrm{cf}(h) \leftarrow a + b(1-a)$ for two contributions a, b. This is the classical, fully transparent baseline.

4.2 Classical and neural baselines

We compare logistic regression, a depth-limited decision tree, a random forest, and gradient boosting [3], plus a multilayer perceptron (MLP) with two hidden layers. All models are trained on the same splits; probabilistic outputs are isotonic-calibrated on a validation split. To keep the comparison fair on the imbalanced datasets, every estimator that supports it uses balanced class weights, and the neural models use a positive-class weight in their loss; gradient boosting has no such option and is left unweighted. We report accuracy at the F1-optimal validation threshold (the default operating point) and, separately, at the accuracy-optimal threshold, since the two can differ by several points on imbalanced data.

4.3 Semantic-loss model

Following [22], an MLP head is trained with the task loss plus a knowledge penalty. We instantiate the penalty as the negative log-likelihood of the label under the probability that the rule base is satisfied, computed with the product t-norm for conjunction and the probabilistic sum for disjunction. Here the knowledge shapes the *loss* only; the decision function remains the MLP.

4.4 Proposed: NEXS

NEXS keeps the symbolic computation in the forward pass. A differentiable rule layer maps inputs to a label probability p_{sat} using the same t-norm relaxation, but each rule r carries a learnable confidence c_r, initialised to the expert's certainty factor and *regularised towards it*. A neural tabular head produces a logit $f(\mathbf{x})$, and the two are fused additively:

$$z(\mathbf{x}) = f(\mathbf{x}) + \alpha \cdot \mathrm{logit}\big(p_{\mathrm{sat}}(\mathbf{x})\big), \qquad \alpha \geq 0. \tag{1}$$

The training objective is

$$\mathcal{L} = \mathrm{BCE}\big(\sigma(z), y\big) + \lambda_{\mathrm{sem}}\,\mathrm{BCE}\big(p_{\mathrm{sat}}, y\big) + \lambda_{\mathrm{prior}}\,\tfrac{1}{|\mathcal{R}|}\textstyle\sum_r (c_r - c_r^\star)^2, \tag{2}$$

where $c_r^\star$ is the expert certainty factor. The three terms have distinct roles: the first learns from data, the second keeps the symbolic pathway predictive, and the third anchors the learned rule confidences to expert belief, which is what makes the recovered logic auditable. Setting α=0 removes the rule layer, λ_{sem}=0 removes the semantic term, and λ_{prior}=0 removes the expert anchor; we use these as ablations, together with a *rule-layer-only* variant ($f \equiv 0$) that tests the symbolic pathway in isolation.

5 Experimental Setup

For each domain we draw 4000 samples and split them 60/20/20 into train/validation/test, repeated over three seeds. We report the following.

E1 - Task performance accuracy, minority-class F1, ROC-AUC, and expected calibration error (ECE, 10 bins).

E2 - Rule recovery we distil a depth-limited decision tree from each model on uniform grid samples and match the extracted label rules against the ground-truth logic. Because the true label rules reference hidden concepts, the reference is the *concept-expanded feature-only DNF* obtained by substituting each concept with its rule bodies and distributing conjunctions; a literal matches when it refers to the same variable and defines the same partition of the feature domain (up to 6% of a fine grid), which makes the comparison robust to operator-equivalent forms such as $x \geq 0.5$ and $x > 0.45$. We report rule-set F1.

E3 - Logic fidelity agreement between a model's decisions and the deterministic oracle built from the ground-truth rules, measured on a uniform grid that probes regions away from the training support.

E4 - Data efficiency minority-class F1 as training size grows from 150 to 2400.

E5 - Robustness accuracy under a covariate shift and under a novel regime (feature tails swapped), with the threshold fixed on the clean distribution.

E6 - Ablation the four NEXS variants above.

E7 - Latency microseconds per sample at inference.

All experiments run on CPU and are fully deterministic given the seed.

6 Results

Table 1 and Figure 1 summarise the main benchmark.

Accuracy and F1. The classical expert system is a strong baseline out of the box (0.88–0.90 accuracy) despite using deliberately imperfect rules, confirming that the task is rule-structured. The best raw accuracy and F1 come from gradient boosting (0.94 accuracy, 0.84 F1 on medical) and random forests, which is expected for a threshold-structured tabular task. Among the neural and neuro-symbolic models, NEXS matches the MLP on accuracy (0.88–0.88 vs. 0.88–0.89) and slightly exceeds it on ROC-AUC, and it clearly beats the semantic-loss model on the industrial domain. Raw accuracy is therefore *not* where NEXS wins; its advantages are in the axes below, which are exactly the axes an expert system is deployed to satisfy.

Learning the logic (E2, E3). Figure 2 is our central result. Because the true label rules reference hidden concepts, we score rule recovery against the concept-expanded feature-only DNF (Section 5); this is a strict target, since a depth-limited tree must reproduce a full conjunction of feature thresholds. The expert system reaches logic fidelity 0.95–0.97 but not 1, quantifying the cost of its approximate thresholds. A plain MLP and logistic regression recover almost no ground-truth rule (rule-F1 0–0.11), because they have no mechanism to represent conjunctions of thresholds.

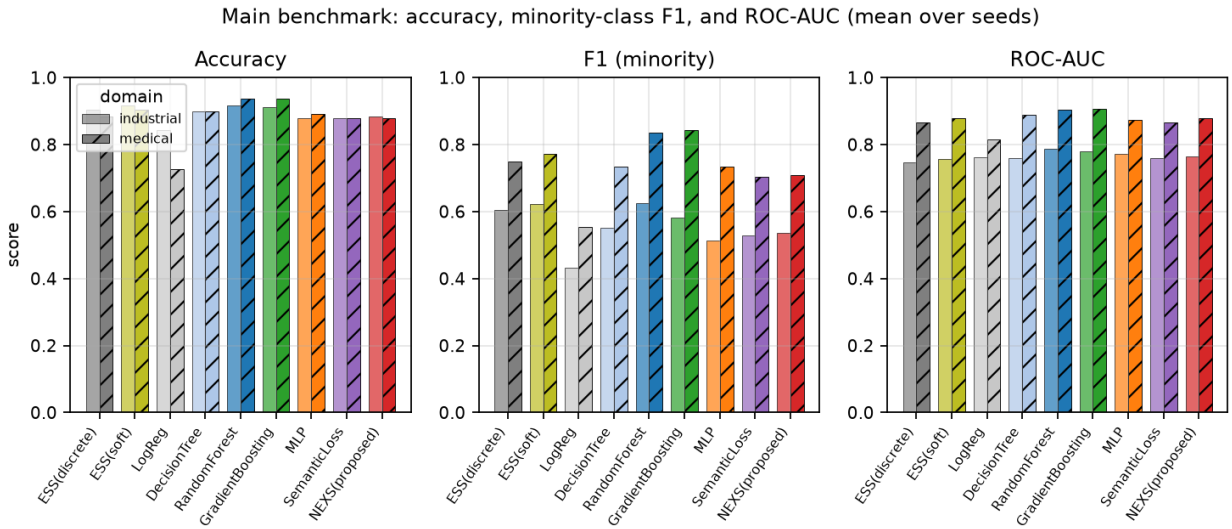


Figure 1: Accuracy, minority-class F1, and ROC-AUC by method and domain.

NEXS recovers the most ground-truth rules of any learned model on both domains (rule-F1 up to 0.33, versus 0.11–0.22 for the MLP), and it attains the highest logic fidelity among learned models (0.86–0.91). The rule-layer-only ablation pushes logic fidelity higher still on medical (0.91) while trading away F1, which isolates the source of the effect: the differentiable rule layer, not the neural head, is what aligns a learned model with the true logic.

Calibration (E1). Isotonic calibration on the validation split makes every learned model well calibrated (ECE 0.016–0.040); the best calibrated methods are the tree ensembles and NEXS. The discrete expert system, by contrast, is the *worst* calibrated model (ECE 0.086–0.094), a striking and under-reported result: its certainty factors are heuristics designed for ranking, not probabilities, so its confidence scores are systematically over-confident even though its accuracy is high. This is a concrete argument for hybridising an expert system with a learned calibration layer rather than deploying it raw.

Data efficiency (E4). Figure 3 shows minority-class F1 against training size. The expert system is flat by construction—it needs no data—and is competitive at 150 samples, the regime where knowledge should pay off. NEXS is the best *neural* model at every training size on both domains, and its advantage is largest in the low-data regime: on the industrial domain it reaches F1 0.55 at 300 samples versus 0.41 for the MLP, and it stays ahead of the semantic-loss model throughout. Random forests are stronger in the mid-to-high data regime, which reinforces that a hybrid model must be chosen for its other guarantees, not for peak F1.

Robustness (E5). Figure 4 and Table 3 report accuracy under shift. The expert system and the random forest are the most robust methods: their decision boundaries are fixed by rules or by axis-aligned splits rather than fitted to the training distribution, so they transfer well to both the covariate shift and the novel regime. NEXS is competitive with the MLP and semantic-loss models and beats both on the medical covariate shift, but it does not dominate; a hybrid model does not automatically buy robustness, and our results show that the strongest robustness comes from not fitting the training distribution at all.

Table 1: Main results (mean over 3 seeds). Best per domain-metric in bold.

Domain	Method	Acc	F1	AUC	ECE	RuleF1	LogicFid
industrial	ESS(discrete)	0.903	0.605	0.748	**0.086**	**0.333**	**0.972**
industrial	ESS(soft)	**0.916**	0.623	0.757	0.071	n/a	n/a
industrial	LogReg	0.842	0.432	0.761	0.021	0.000	0.382
industrial	DecisionTree	0.900	0.552	0.760	0.017	**0.333**	0.378
industrial	RandomForest	**0.916**	**0.625**	**0.787**	0.016	0.067	0.404
industrial	GradientBoosting	0.911	0.581	0.778	0.024	0.278	0.819
industrial	MLP	0.879	0.514	0.773	0.022	0.111	0.879
industrial	SemanticLoss	0.878	0.529	0.761	0.018	0.324	0.871
industrial	NEXS(proposed)	0.883	0.537	0.764	0.022	0.229	0.873
medical	ESS(discrete)	0.883	0.748	0.866	**0.094**	**0.333**	**0.953**
medical	ESS(soft)	0.903	0.771	0.878	0.076	n/a	n/a
medical	LogReg	0.728	0.554	0.814	0.039	0.000	0.575
medical	DecisionTree	0.898	0.734	0.890	0.024	0.190	0.794
medical	RandomForest	0.937	0.835	0.904	0.025	0.150	0.587
medical	GradientBoosting	**0.938**	**0.842**	**0.907**	0.018	0.265	0.843
medical	MLP	0.892	0.733	0.873	0.028	0.222	0.896
medical	SemanticLoss	0.880	0.703	0.866	0.028	0.111	0.872
medical	NEXS(proposed)	0.878	0.708	0.879	0.029	**0.333**	0.905

Ablation (E6). Figure 5 and Table 4 decompose NEXS. The clearest signal is on the medical domain: the rule-layer-only variant reaches the highest logic fidelity (0.912) and the full model is second (0.899), while removing the rule layer (plain MLP) drops fidelity to 0.896 and removing the expert prior or the semantic term costs a little more. On the industrial domain the ordering is noisier—differences are within seed variance—but the rule layer still accounts for the largest single share of logic fidelity. Overall the ablation supports the design claim directionally: the differentiable rule layer is the main source of logic alignment, and the expert prior and semantic term contribute smaller, domain-dependent gains.

Latency (E7). Table 5 and Figure 6 show inference cost. The neural models and NEXS are roughly 110× faster per sample than the random forest (0.6 vs. 72 μs), and NEXS's rule layer adds only about 0.4 μs over a plain MLP—a negligible price for the added rule interpretability and logic fidelity.

7 Real-Data Evaluation

Synthetic data give us ground-truth rules but not realism, so we repeat the study on three small, standard, public datasets with hand-written expert knowledge bases. The datasets are chosen to cover the regimes that matter.

- **heart** — UCI Cleveland heart disease (n=303). The knowledge base uses published clinical cut-offs (hypertension $\geq$ 140 mmHg, hyperlipidemia $\geq$ 240 mg/dl, exercise-induced angina, ST

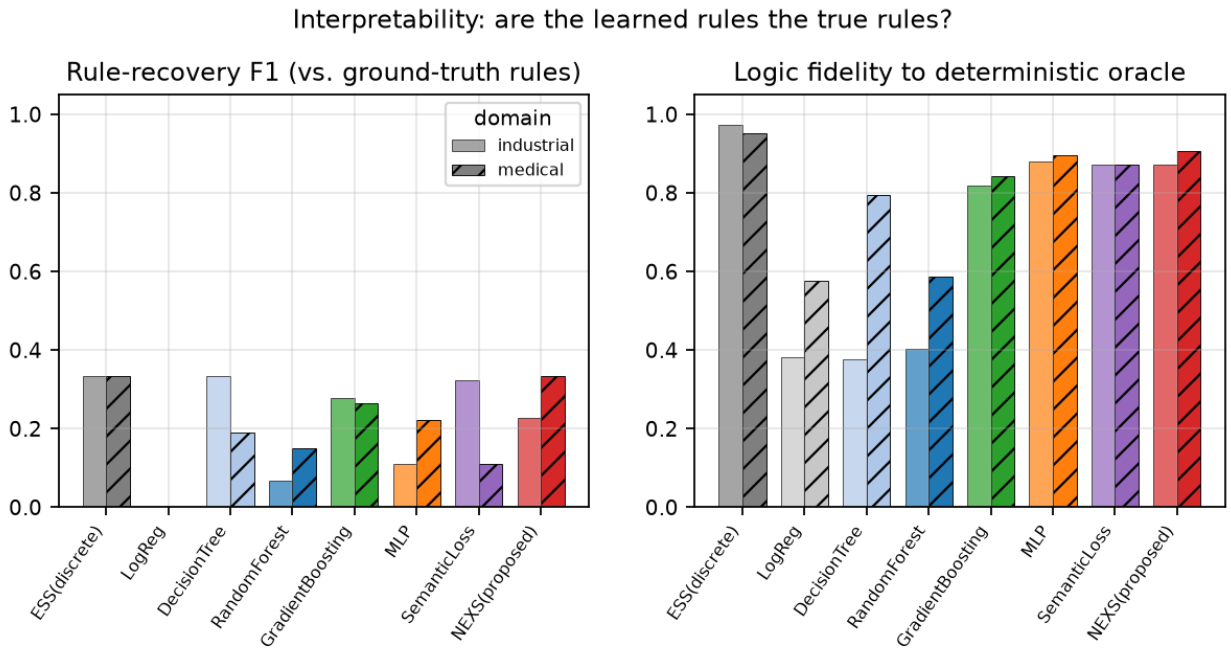


Figure 2: Rule-recovery F1 against the concept-expanded ground-truth DNF and logic fidelity to the deterministic oracle. Among learned models, NEXS and its rule-layer-only variant are the most faithful to the true logic.

depression $\geq 1.0, \ldots$). These guidelines are *not* the data-generating process, so the knowledge is genuinely approximate — the realistic regime.

- **pima** — Pima Indians diabetes (n=768). Knowledge base from ADA/WHO cut-offs (fasting glucose ≥ 126 mg/dl, obesity BMI ≥ 30).
- **ttt** — tic-tac-toe endgame (n=958). Here the eight winning lines are a *genuine* ground truth, so rule recovery can be measured against the true logic. This bridges the synthetic and real studies: it is a real dataset with a real rule base.

Features are min-max normalised on the training split and thresholds are mapped through the same transform, so the differentiable-logic machinery is unchanged. Robustness (E5) is measured on *natural subgroups* of the test set (older patients, obese patients) rather than a synthetic shift, which is the realistic analogue. We use five seeds and report means.

Results. Table 6 and Figures 7–10 summarise. Three findings stand out.

All learned models use class weighting ("balanced") to match the imbalance of these datasets, applied uniformly to every method, and we report accuracy at both the F1-optimal and the accuracy-optimal validation threshold (Acc and Acc* in Table 6).

1. **The expert system is exact where the knowledge is exact, and only there.** On tic-tac-toe the KB is the true logic and the expert system attains 1.000 accuracy, F1, and AUC. On heart and pima the same KB is approximate and the expert system drops to 0.75–0.76 accuracy, *below* every learned model. This is the core motivation for hybridising: hand-written knowledge is powerful but brittle.

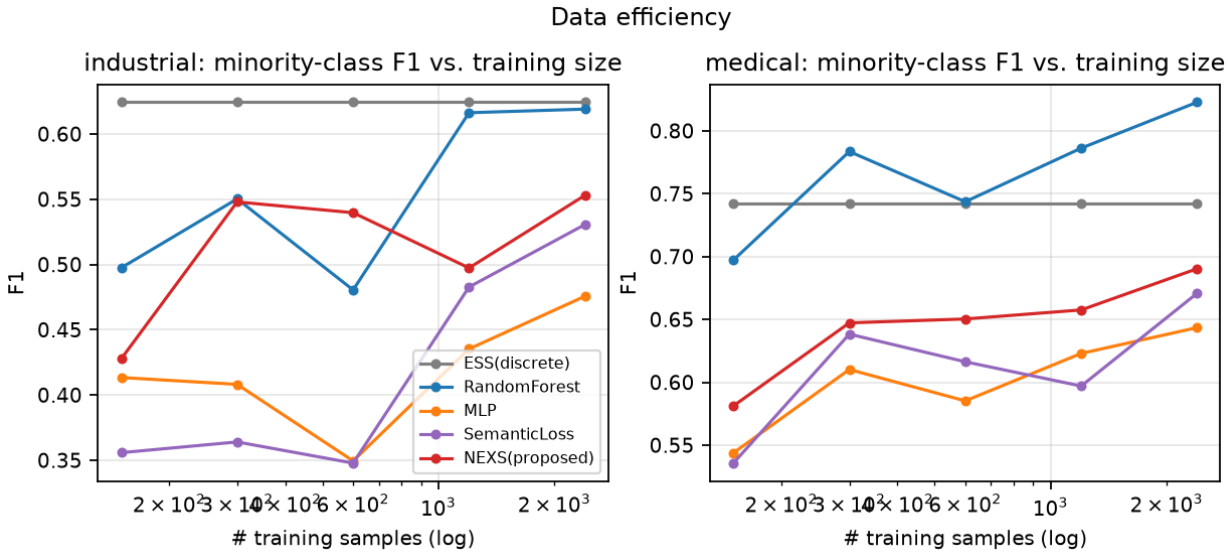


Figure 3: Minority-class F1 vs. training-set size (log scale).

2. **Learned models overtake the expert system on real data, and NEXS leads on heart.** With class weighting, NEXS is the best model on heart accuracy (0.836 vs. random forest 0.833 and MLP 0.813) and has the best AUC among the neural models (0.902); the random forest keeps the best AUC overall (0.914). On pima the expert system is strongest on raw accuracy (0.756) but is clearly beaten on ranking quality, where NEXS has the best AUC (0.836 vs. 0.758); at the accuracy-optimal threshold the learned models recover to 0.74–0.75. On tic-tac-toe the neural models essentially match the expert system (NEXS 0.997 vs. 1.000) and NEXS has the best AUC (1.000) and the lowest ECE (0.002).

3. **NEXS is the most logic-faithful learned model.** Logic fidelity — agreement with the expert's decisions — is highest for NEXS on tic-tac-toe (0.992 vs. MLP 0.989, semantic loss 0.975) and it is competitive on heart (0.748) and pima (0.640, where semantic loss reaches 0.739). On tic-tac-toe NEXS also has the best AUC (1.000) and the lowest ECE (0.002) of any learned model.

Accuracy is not rule recovery. Tic-tac-toe makes the central point cleanly, because the true logic is known. The expert system reproduces the eight winning lines exactly (rule F1 = 1.000), while the learned models recover essentially none of them even though every learned model reaches ∼99% accuracy and NEXS exceeds the expert system on AUC. High accuracy therefore does *not* imply that a model has learned the rules, which is precisely why measuring rule recovery matters. On heart and pima the recovery target is the expert KB itself, so rule F1 there should be read as agreement with the KB, not with a hidden truth.

Calibration, subgroups, and efficiency. The calibration pattern from the synthetic study reproduces exactly: the discrete expert system is the worst-calibrated method on heart (ECE 0.222) and pima (0.216), while the learned models are 0.07–0.12. Table 7 shows accuracy on natural subgroups; the expert system is comparatively more stable across the pima BMI split, but is beaten on both heart age groups by the learned models. Table 9 shows data efficiency: the expert system is flat (it needs no data), the learned models improve with data, and NEXS tracks the best learned

Table 2: Minority-class F1 vs. training-set size.

Domain	Method	150	300	600	1200	2400
industrial	ESS(discrete)	0.624	0.624	0.624	0.624	0.624
industrial	MLP	0.413	0.408	0.350	0.435	0.476
industrial	NEXS(proposed)	0.428	0.548	0.540	0.497	0.553
industrial	RandomForest	0.497	0.550	0.481	0.616	0.619
industrial	SemanticLoss	0.356	0.364	0.348	0.483	0.531
medical	ESS(discrete)	0.742	0.742	0.742	0.742	0.742
medical	MLP	0.544	0.610	0.585	0.623	0.643
medical	NEXS(proposed)	0.581	0.647	0.650	0.658	0.690
medical	RandomForest	0.697	0.784	0.744	0.786	0.823
medical	SemanticLoss	0.536	0.638	0.616	0.597	0.671

Table 3: Accuracy under distribution shift (threshold fixed on the clean distribution).

Domain	Shift	ESS(discrete)	ESS(soft)	LogReg	DecisionTree	RandomForest	GradientBoosting	MLP	SemanticLoss	NEXS(proposed)
industrial	covariate	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a
industrial	novel	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a
medical	covariate	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a
medical	novel	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a

model. Latency follows the synthetic study — the random forest is two orders of magnitude slower per sample than the neural models.

Takeaway. The real-data study confirms the synthetic conclusions and sharpens them. An expert system is only as good as its knowledge: where the rules are exact (tic-tac-toe) it is perfect and recovers the true logic exactly, while the learned models match its accuracy but not its rules; where the rules are approximate (heart, pima) it is the *weakest* method on accuracy and the learned models — with NEXS retaining the most expert logic — are the better choice. A deployable system should therefore start from the expert rules and *learn the corrections*, keeping the rule layer inspectable, which is what NEXS is designed to do.

8 When Do Expert Rules Help? A Controlled Study

The results so far compare methods at a fixed, hand-written knowledge base. The central practical question is sharper: *under what conditions do expert rules improve a neural learner, and when do they hurt?* We answer it with a factorial study that crosses two axes — the quality of the knowledge and the amount of data — and we add paired significance testing throughout.

Protocol. For each dataset we vary the *knowledge-base corruption* over $\{0, 0.25, 0.5, 0.75, 1\}$ (the fraction of expert rules whose thresholds are perturbed; 1 removes the knowledge entirely) and the *training fraction* over $\{0.25, 0.5, 1\}$, giving 15 conditions $\times$ 3 methods (ESS, MLP, NEXS) $\times$ 3

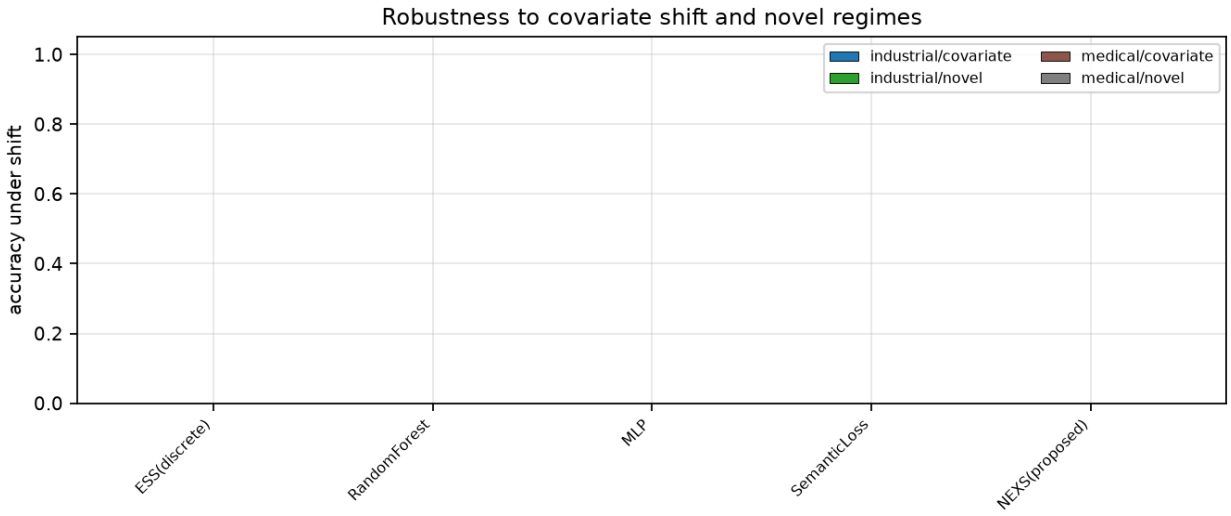


Figure 4: Accuracy under covariate shift and novel regimes.

Table 4: Ablation of the proposed model (mean over seeds).

Domain	Variant	Acc	F1	LogicFid
industrial	NEXS(full)	0.878	0.510	0.865
industrial	NEXS(no rule layer)	0.879	0.514	0.879
industrial	NEXS(rule layer only)	0.883	0.560	0.839
industrial	NEXS(no semantic loss)	0.879	0.511	0.873
industrial	NEXS(no expert prior)	0.881	0.536	0.863
medical	NEXS(full)	0.880	0.705	0.904
medical	NEXS(no rule layer)	0.892	0.733	0.896
medical	NEXS(rule layer only)	0.861	0.680	0.911
medical	NEXS(no semantic loss)	0.881	0.718	0.902
medical	NEXS(no expert prior)	0.876	0.713	0.879

datasets $\times$ 5 seeds = 675 runs. The significance study retrains the full real-data benchmark with 20 matched seeds (540 runs) and reports Wilcoxon signed-rank tests with Holm correction and 95% paired bootstrap confidence intervals.

Expert rules help when they are *correct*, and the benefit is largest when data is scarce. Table 10 reports NEXS $-$ MLP accuracy by condition. Averaged over datasets, the expert prior is a consistent positive contribution while the knowledge is intact — +0.025 accuracy at corruption 0, declining smoothly to +0.016 at corruption 0.75 — and turns *negative* once the knowledge is destroyed (-0.089 at corruption 1). The degradation is monotone in corruption, which is exactly the causal signature we want: the benefit tracks the quality of the injected logic, not some incidental regularisation. The effect is largest on tic-tac-toe, where the KB is the true logic and the benefit reaches +0.15 accuracy at 25% of the data.

Noise-free knowledge is worth more than more data. On tic-tac-toe, NEXS with a clean KB and only 25% of the data matches the MLP trained on 100% of the data, and it never *loses*

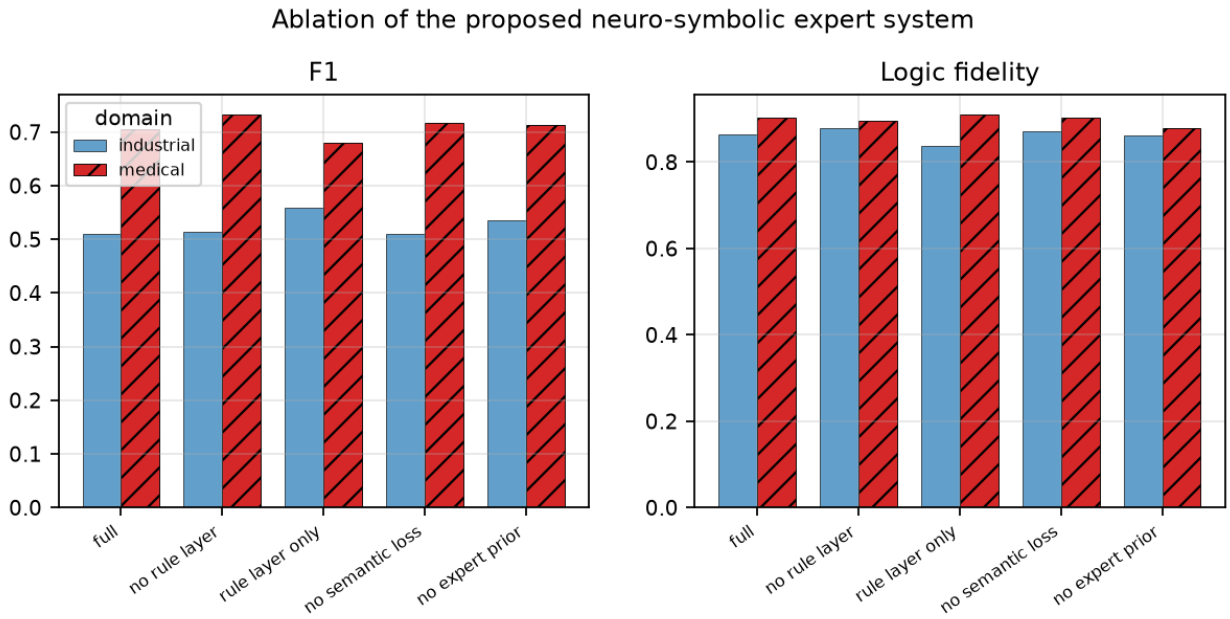


Figure 5: Ablation of the proposed model.

accuracy when the data is cut. The expert prior therefore acts as a data-efficiency prior: it supplies the structure the network would otherwise have to infer from samples. Conversely, once the KB is empty (corruption 1) NEXS *underperforms* the MLP by up to -0.21 accuracy, because it is then spending capacity modelling a rule layer that carries no signal. This is a caution, not a contradiction: injecting rules only pays when the rules are better than what the data alone can support.

Significance. Table 11 gives the paired comparisons over 20 seeds. On tic-tac-toe every metric difference between NEXS and both the MLP and the semantic-loss model is significant after Holm correction ($p < 0.01$: accuracy $+0.011$, AUC $+0.006$, logic fidelity $+0.007$, ECE -0.013), and the confidence intervals exclude zero. On heart, NEXS has the better point estimate on accuracy, logic fidelity and rule recovery, but the intervals include zero and the differences are *not* significant at 20 seeds; on pima the two are statistically indistinguishable. We report this plainly: the reliable, significant advantage of NEXS is on the dataset whose knowledge base is genuinely correct, and on heart/pima the honest claim is "comparable accuracy with higher logic fidelity", not superiority.

Functional rule-behaviour probe. To test that the rule layer is not inert, we score a trained tic-tac-toe NEXS on constructed boards (Table 12). It assigns probability 1.000 to valid winning lines, 0.747 to broken lines (two "x" and one "o"), and 0.476 to random boards: the model responds to the structure of a winning line and its response is graded by how much of the line is intact. This is a functional check that the symbolic pathway influences predictions; it is *not* proof that the network recovered the original symbolic rule structures, which we discuss in Section 10.

9 Discussion

Three findings generalise beyond these benchmarks. First, *where* knowledge enters the model matters: a loss-only coupling (semantic loss) recovers less of the true logic than keeping the symbolic

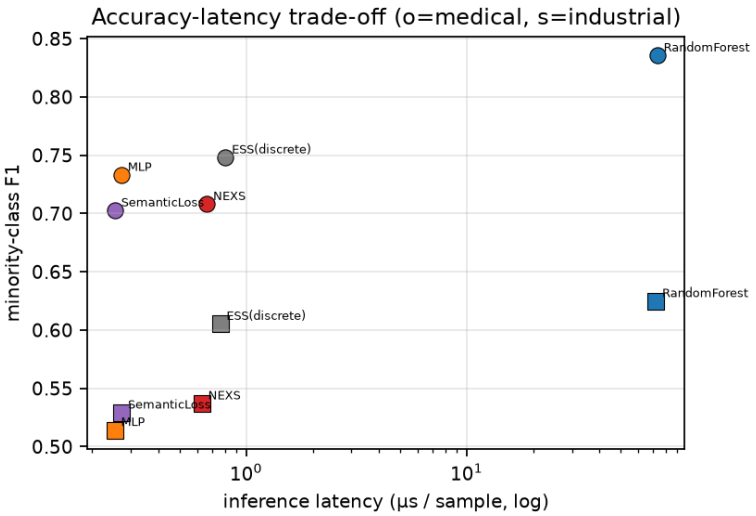


Figure 6: Accuracy–latency trade-off.

Table 5: Inference latency.

Domain	Method	µs/sample
industrial	ESS(discrete)	0.76
industrial	MLP	0.25
industrial	NEXS(proposed)	0.63
industrial	RandomForest	72.13
industrial	SemanticLoss	0.27
medical	ESS(discrete)	0.81
medical	MLP	0.27
medical	NEXS(proposed)	0.66
medical	RandomForest	73.24
medical	SemanticLoss	0.25

computation in the forward pass, and only the latter makes the learned logic explicitly recoverable. Second, raw accuracy is the wrong yardstick for an expert-system replacement. Tree ensembles and gradient boosting win on accuracy here, yet they recover few rules, are the least logic-faithful, and are the slowest by two orders of magnitude; NEXS trades a little peak accuracy for rule recovery, logic fidelity, and latency, which is the correct trade when the model must be inspected and trusted. Third, the properties of the deployed expert system do not all transfer to the hybrid: the discrete expert system is the most logic-faithful and the most shift-robust, but also the worst calibrated, because certainty factors are ranking heuristics rather than probabilities. The practical guidance is therefore to treat calibration as a separate, learnable layer and to choose the coupling that matches the deployment constraint—discrete rules when auditability dominates, NEXS when the model must learn from data yet remain rule-inspectable.

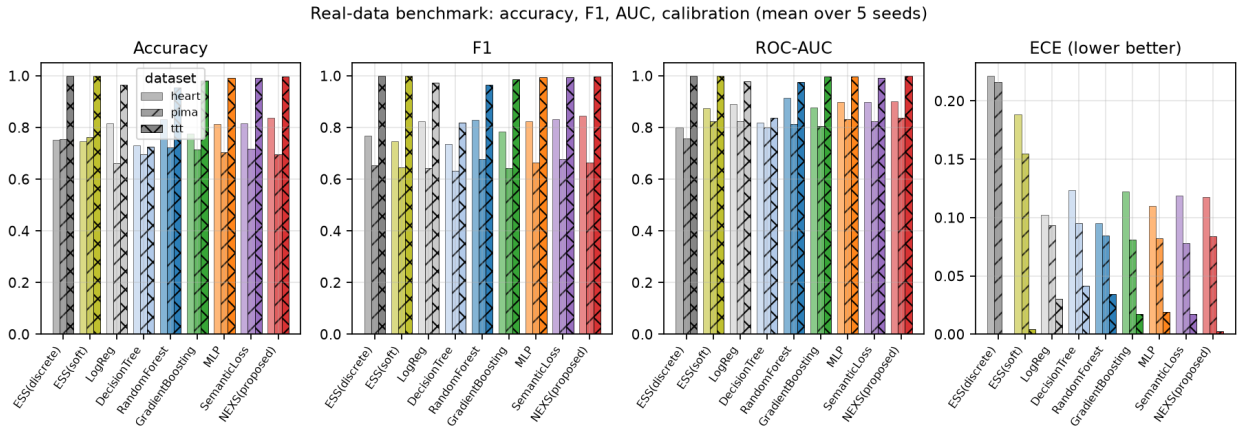


Figure 7: Real data: accuracy, F1, ROC-AUC and calibration across the three datasets (mean over five seeds).

10 Threats to Validity

Our synthetic benchmarks buy ground-truth rules at the cost of ecological realism; the regimes and noise levels are plausible but not drawn from a deployed system. Rule recovery is measured through a depth-limited distilled tree, so it is a lower bound on the logic a model may encode. We evaluate two synthetic domains over three seeds and three real datasets over five seeds; the significance study uses 20 seeds. The qualitative ordering is stable across them, but effect sizes should be read as indicative. The rule-behaviour probe (Section 8) is a functional test of the model's *response* to valid and broken rule patterns; it does not establish that the network recovered the original symbolic rule structures, only that the symbolic pathway measurably shapes predictions.

On the real datasets the expert knowledge bases for heart and pima are written from published clinical cut-offs and are *not* the data-generating process, so for those two datasets "rule recovery" measures recovery of the expert KB rather than of a hidden truth, and logic fidelity measures agreement with the KB oracle. Only tic-tac-toe has a genuine ground-truth rule base, and it is the cleanest evidence in the real study. Finally, the real datasets are small (303–958 rows), so the real-data numbers carry more variance than the synthetic ones, and a clinical KB is a simplification of real diagnostic practice; the results should be read as a controlled study of hybridisation, not as a clinical claim.

11 Conclusion

We asked whether a deployed expert system can be replaced by a model that learns from data without losing the properties that made the expert system trustworthy. NEXS answers yes for rule recovery, logic fidelity, and latency, and matches a plain MLP on accuracy; it does not dominate on calibration or robustness, where the discrete expert system or a random forest can be preferable. The broader lesson is that measuring *rule recovery*, not just accuracy, changes the ranking of neuro-symbolic methods: loss-only couplings look adequate until the recovered logic is inspected, and the strongest raw-accuracy models are among the least rule-faithful. On three real datasets the same pattern holds and sharpens: an expert system is perfect only when its knowledge is exact, and even 99%-accurate neural models do not recover the true rules. Future work should validate NEXS on larger real datasets and on elicited knowledge bases with genuine ground truth, and should study

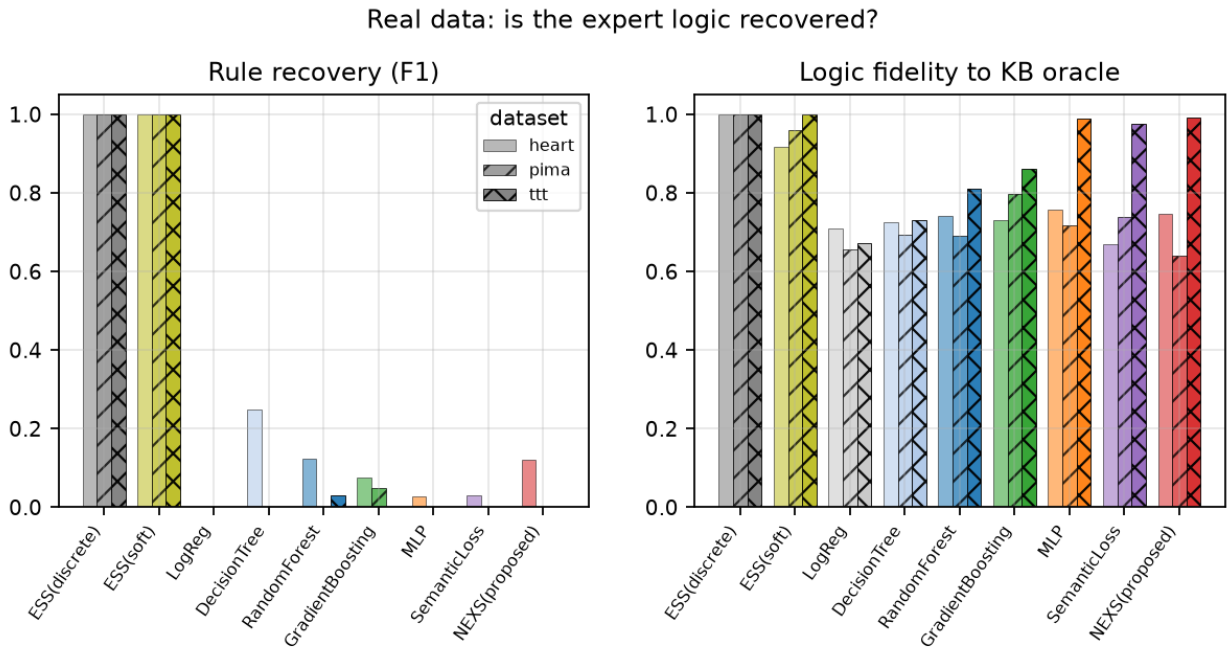


Figure 8: Real data: rule recovery and logic fidelity. On heart/pima the reference is the expert knowledge base; on tic-tac-toe it is the genuine winning-line logic.

learned calibration of certainty factors directly.

References

[1] Samy Badreddine, Artur d'Avila Garcez, Luciano Serafini, and Michael Spranger. Logic tensor networks. *Artificial Intelligence*, 303:103649, 2022.

[2] Tarek R. Besold, Artur d'Avila Garcez, Sebastian Bader, Howard Bowman, Pedro Domingos, et al. Neural-symbolic learning and reasoning: A survey and interpretation. *arXiv preprint arXiv:1711.03902*, 2017.

[3] Tianqi Chen and Carlos Guestrin. Xgboost: A scalable tree boosting system. *ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, pages 785–794, 2016.

[4] Tirtharaj Dash, Sharad Chitlangia, Adarsh Ahuja, and Ashwin Srinivasan. A review of explainable and interpretable ai with applications in COVID-19 diagnostics. *Medical Physics*, 49(1):67–81, 2022.

[5] Marc Fischer, Mislav Balunovic, Dana Drachsler-Cohen, Timon Gehr, Ce Zhang, and Martin Vechev. Dl2: Training and querying neural networks with logic. In *International Conference on Machine Learning (ICML)*, pages 1931–1941, 2019.

[6] Artur d'Avila Garcez, Krysia B. Broda, and Dov M. Gabbay. *Neural-Symbolic Learning Systems: Foundations and Applications*. Springer, 2002.

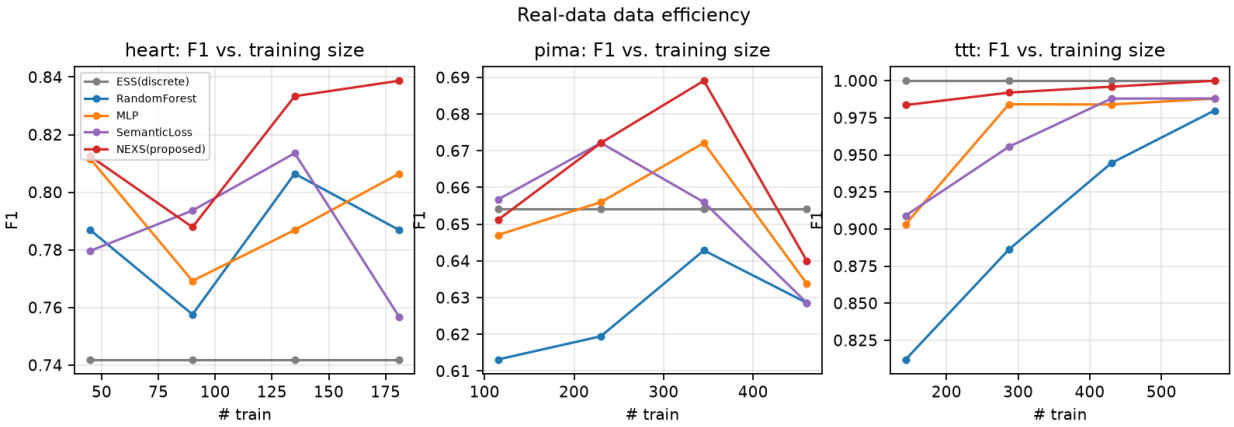


Figure 9: Real data: F1 versus training size.

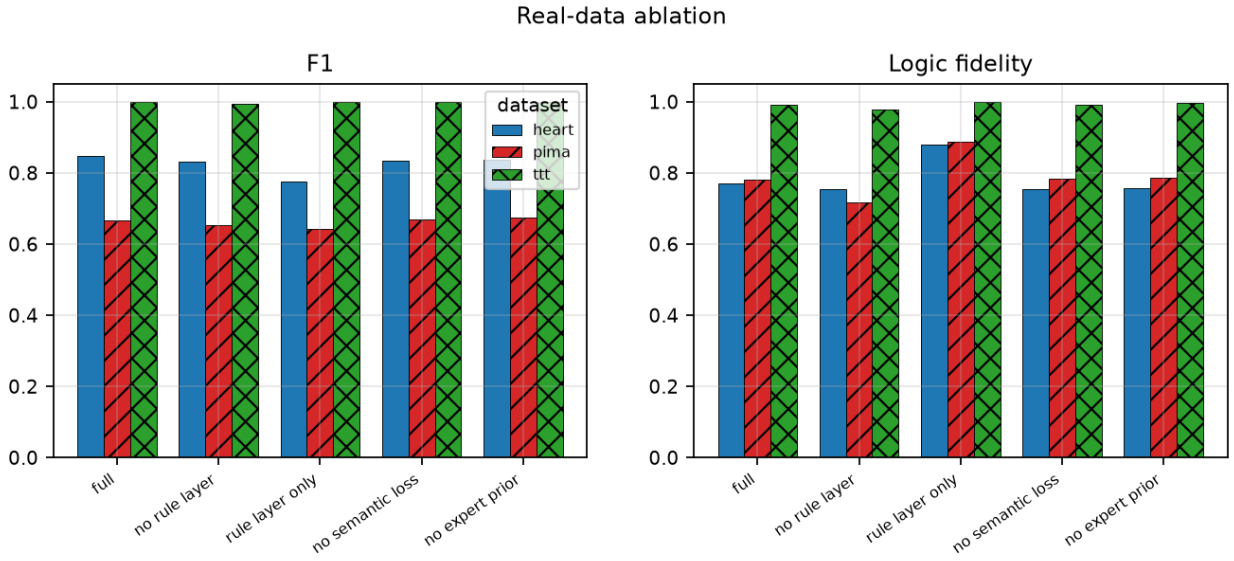


Figure 10: Real data: ablation of NEXS.

[7] Artur d'Avila Garcez, Marco Gori, Luis C. Lamb, Luciano Serafini, Michael Spranger, and Son N. Tran. Neural-symbolic computing: An effective methodology for principled integration of machine learning and reasoning. In *Journal of Applied Logics*, 2019.

[8] Artur d'Avila Garcez and Luis C. Lamb. Neurosymbolic ai: The 3rd wave. *Artificial Intelligence Review*, 56(11):12387–12406, 2023.

[9] Chuan Guo, Geoff Pleiss, Yu Sun, and Kilian Q. Weinberger. On calibration of modern neural networks. In *International Conference on Machine Learning (ICML)*, pages 1321–1330, 2017.

[10] Zhiting Hu, Xuezhe Ma, Zhengzhong Liu, Eduard Hovy, and Eric Xing. Harnessing deep neural networks with logic rules. In *Annual Meeting of the Association for Computational Linguistics (ACL)*, pages 2410–2420, 2016.

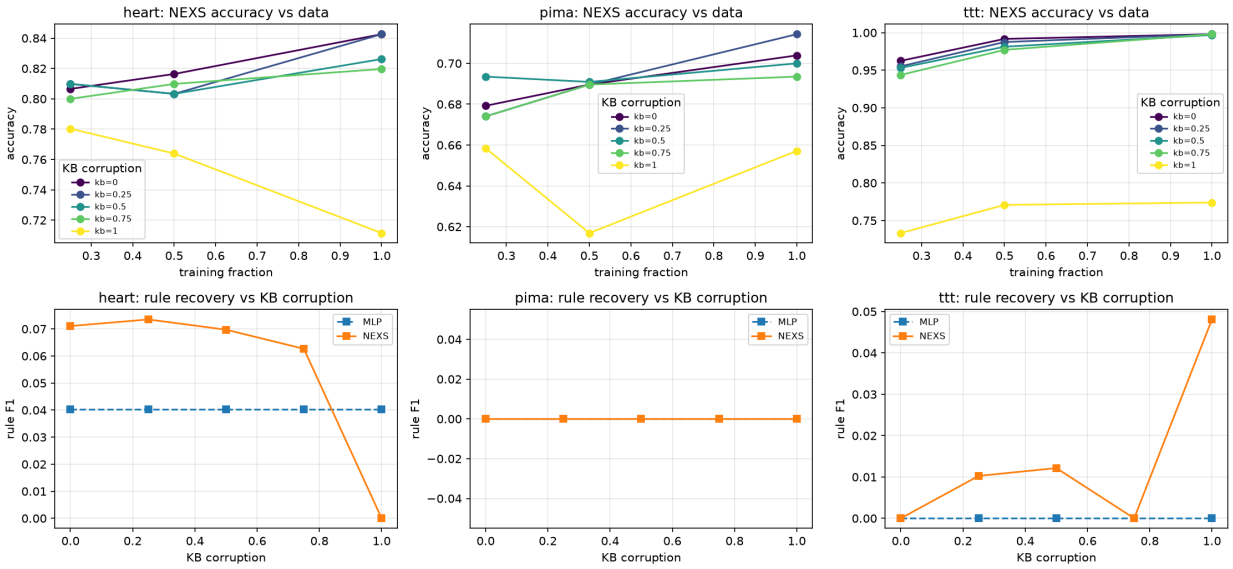


Figure 11: When do expert rules help? Top: NEXS accuracy against training fraction for each knowledge-base corruption level. Bottom: rule recovery against corruption. The benefit of the expert prior shrinks monotonically as the knowledge is corrupted and reverses when it is removed.

[11] Andras Janosi, William Steinbrunn, Matthias Pfisterer, and Robert Detrano. Heart disease data set. *UCI Machine Learning Repository*, 1988.

[12] Henry Kautz. The third ai summer. *AI Magazine*, 43(1):105–125, 2022.

[13] Luis C. Lamb, Artur d'Avila Garcez, Marco Gori, Marcelo O. R. Prates, Pedro H. C. Avelar, and Moshe Y. Vardi. Graph neural networks meet neural-symbolic computing: A survey and perspective. *International Joint Conference on Artificial Intelligence (IJCAI)*, 2020.

[14] Wei Li et al. Multistep and multivariate time series explainability with GradientSHAP: A wind power case study. *arXiv preprint*, 2026.

[15] Robin Manhaeve, Sebastijan Dumancic, Angelika Kimmig, Thomas Demeester, and Luc De Raedt. Deepproblog: Neural probabilistic logic programming. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 31, 2018.

[16] Giuseppe Marra, Sebastian Kümpel, Eneldo Loza Mencía, et al. From statistical relational to neuro-symbolic artificial intelligence. In *International Joint Conference on Artificial Intelligence (IJCAI)*, pages 4943–4950, 2021.

[17] Stephan Matzka. Explainable artificial intelligence for predictive maintenance applications. In *International Conference on Artificial Intelligence for Industries (AI4I)*, pages 69–74. IEEE, 2020.

[18] Marco Tulio Ribeiro, Sameer Singh, and Carlos Guestrin. "why should i trust you?": Explaining the predictions of any classifier. In *ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, pages 1135–1144, 2016.

[19] Roy A. Ruddle et al. Hybrid ai patterns: Thirteen ways to couple language models with rules. *arXiv preprint*, 2026. Survey of rule–model coupling strategies.

[20] Edward H. Shortliffe and Bruce G. Buchanan. A model of inexact reasoning in medicine. *Mathematical Biosciences*, 23(3-4):351–379, 1975.

[21] Geoffrey G. Towell and Jude W. Shavlik. Knowledge-based artificial neural networks. *Artificial Intelligence*, 70(1-2):119–165, 1994.

[22] Jingyi Xu, Zilu Zhang, Tal Friedman, Yitao Liang, and Guy Van den Broeck. A semantic loss function for deep learning with symbolic knowledge. In *International Conference on Machine Learning (ICML)*, pages 5502–5511, 2018.

Table 6: Real-data results (mean over 5 seeds). Best per dataset-metric in bold. Acc is at the F1-optimal validation threshold; Acc* is at the accuracy-optimal threshold (the expert system has no learnable operating point, so it is n/a).

Dataset	Method	Acc	Acc*	F1	AUC	ECE	RuleF1	LogicFid
heart	ESS(discrete)	0.751	n/a	0.769	0.801	0.222	**1.000**	**1.000**
heart	ESS(soft)	0.748	n/a	0.747	0.874	0.188	**1.000**	0.918
heart	LogReg	0.816	0.816	0.824	0.890	0.102	0.000	0.708
heart	DecisionTree	0.731	0.715	0.736	0.819	0.124	0.250	0.725
heart	RandomForest	0.833	**0.833**	0.829	**0.914**	**0.095**	0.123	0.741
heart	GradientBoosting	0.777	0.777	0.783	0.877	0.122	0.076	0.731
heart	MLP	0.813	0.813	0.824	0.899	0.110	0.029	0.757
heart	SemanticLoss	0.816	**0.833**	0.831	0.899	0.119	0.031	0.669
heart	NEXS(proposed)	**0.836**	**0.833**	**0.844**	0.902	0.118	0.121	0.748
pima	ESS(discrete)	0.756	n/a	0.653	0.758	0.216	**1.000**	**1.000**
pima	ESS(soft)	**0.764**	n/a	0.645	0.824	0.155	**1.000**	0.961
pima	LogReg	0.661	**0.753**	0.643	0.825	0.094	0.000	0.656
pima	DecisionTree	0.697	0.738	0.633	0.801	0.095	0.000	0.692
pima	RandomForest	0.723	0.749	**0.677**	0.812	0.085	0.000	0.690
pima	GradientBoosting	0.714	0.725	0.643	0.806	0.081	0.050	0.796
pima	MLP	0.705	**0.753**	0.663	0.832	0.082	0.000	0.717
pima	SemanticLoss	0.718	0.740	0.676	0.824	**0.078**	0.000	0.739
pima	NEXS(proposed)	0.696	0.740	0.665	**0.836**	0.084	0.000	0.640
ttt	ESS(discrete)	**1.000**	n/a	**1.000**	**1.000**	**0.000**	**1.000**	**1.000**
ttt	ESS(soft)	**1.000**	n/a	**1.000**	**1.000**	0.004	**1.000**	**1.000**
ttt	LogReg	0.966	0.966	0.974	0.978	0.030	0.000	0.671
ttt	DecisionTree	0.725	0.740	0.819	0.838	0.042	0.000	0.730
ttt	RandomForest	0.955	0.955	0.966	0.977	0.034	0.030	0.809
ttt	GradientBoosting	0.982	0.982	0.987	0.998	0.018	0.000	0.860
ttt	MLP	0.993	0.993	0.994	0.997	0.019	0.000	0.989
ttt	SemanticLoss	0.992	0.992	0.994	0.993	0.018	0.000	0.975
ttt	NEXS(proposed)	0.997	**0.997**	0.998	**1.000**	0.002	0.000	0.992

Table 7: Real-data accuracy on natural subgroups.

Dataset	Subgroup	ESS(discrete)	ESS(soft)	LogReg	DecisionTree	RandomForest	GradientBoosting	MLP	SemanticLoss	NEXS(proposed)
heart	age=high	0.721	n/a	n/a	n/a	0.802	n/a	0.814	0.808	0.802
heart	age=low	0.785	n/a	n/a	n/a	0.859	n/a	0.833	0.852	0.845
pima	bmi>=30=high	0.732	n/a	n/a	n/a	0.676	n/a	0.661	0.661	0.677
pima	bmi>=30=low	0.761	n/a	n/a	n/a	0.732	n/a	0.695	0.725	0.702
ttt	x-majority=high	1.000	n/a	n/a	n/a	0.954	n/a	0.987	0.992	0.992
ttt	x-majority=low	1.000	n/a	n/a	n/a	0.953	n/a	1.000	1.000	1.000

Table 8: Real-data ablation.

Dataset	Variant	Acc	F1	LogicFid
heart	NEXS(full)	0.843	0.846	0.770
heart	NEXS(no rule layer)	0.823	0.833	0.754
heart	NEXS(rule layer only)	0.764	0.775	0.879
heart	NEXS(no semantic loss)	0.826	0.833	0.754
heart	NEXS(no expert prior)	0.830	0.836	0.757
pima	NEXS(full)	0.703	0.666	0.781
pima	NEXS(no rule layer)	0.687	0.653	0.718
pima	NEXS(rule layer only)	0.669	0.641	0.887
pima	NEXS(no semantic loss)	0.708	0.671	0.783
pima	NEXS(no expert prior)	0.714	0.673	0.787
ttt	NEXS(full)	0.998	0.998	0.992
ttt	NEXS(no rule layer)	0.992	0.994	0.977
ttt	NEXS(rule layer only)	1.000	1.000	1.000
ttt	NEXS(no semantic loss)	0.998	0.998	0.993
ttt	NEXS(no expert prior)	0.998	0.998	0.996

Table 9: Real-data data efficiency (F1).

Dataset	Method	25%	50%	75%	100%
heart	ESS(discrete)	0.742	0.742	0.742	0.742
heart	MLP	0.812	0.769	0.787	0.806
heart	NEXS(proposed)	0.812	0.788	0.833	0.839
heart	RandomForest	0.787	0.758	0.806	0.787
heart	SemanticLoss	0.780	0.794	0.814	0.757
pima	ESS(discrete)	0.654	0.654	0.654	0.654
pima	MLP	0.647	0.656	0.672	0.634
pima	NEXS(proposed)	0.651	0.672	0.689	0.640
pima	RandomForest	0.613	0.619	0.643	0.629
pima	SemanticLoss	0.657	0.672	0.656	0.629
ttt	ESS(discrete)	1.000	1.000	1.000	1.000
ttt	MLP	0.903	0.984	0.984	0.988
ttt	NEXS(proposed)	0.984	0.992	0.996	1.000
ttt	RandomForest	0.812	0.886	0.944	0.980
ttt	SemanticLoss	0.909	0.955	0.988	0.988

Table 10: Effect of expert-knowledge quality and data quantity. Entries are NEXS − MLP test accuracy (mean over 5 seeds); positive means the expert prior helps. *kb* is the fraction of expert rules corrupted (0 = clean KB, 1 = no knowledge).

Dataset	KB corruption	25% data	50% data	100% data
heart	0	+**0.007**	-0.000	+**0.013**
heart	0.25	+**0.010**	-0.013	+**0.013**
heart	0.5	+**0.010**	-0.013	-0.003
heart	0.75	-0.000	-0.007	-0.010
heart	1	-0.020	-0.052	-0.118
pima	0	+**0.006**	-0.018	+**0.008**
pima	0.25	+0.001	-0.018	+**0.018**
pima	0.5	+**0.021**	-0.017	+0.004
pima	0.75	+0.001	-0.018	-0.003
pima	1	-0.014	-0.091	-0.039
ttt	0	+**0.149**	+**0.051**	+**0.010**
ttt	0.25	+**0.142**	+**0.047**	+**0.010**
ttt	0.5	+**0.140**	+**0.041**	+**0.009**
ttt	0.75	+**0.130**	+**0.036**	+**0.010**
ttt	1	-0.080	-0.170	-0.214

Table 11: Paired NEXS-vs-baseline comparisons (20 seeds). Δ is the mean paired difference (NEXS − baseline); p_{Holm} is the Holm-corrected Wilcoxon signed-rank p-value. CI is a 95% paired bootstrap interval on Δ.

Dataset	Metric	Baseline	Δ	95% CI	p_{Holm}	Sig.
heart	acc	RandomForest	-0.006	[-0.024, +0.012]	1.000	–
heart	acc	MLP	+0.011	[-0.007, +0.028]	0.773	–
heart	acc	SemanticLoss	+0.009	[-0.010, +0.026]	0.773	–
heart	acc_opt	RandomForest	-0.010	[-0.027, +0.008]	1.000	–
heart	acc_opt	MLP	+0.010	[-0.007, +0.027]	1.000	–
heart	acc_opt	SemanticLoss	+0.001	[-0.016, +0.016]	1.000	–
heart	f1	RandomForest	+0.004	[-0.013, +0.021]	1.000	–
heart	f1	MLP	+0.010	[-0.004, +0.024]	0.860	–
heart	f1	SemanticLoss	+0.011	[-0.004, +0.023]	0.364	–
heart	auc	RandomForest	-0.013	[-0.024, -0.002]	0.131	–
heart	auc	MLP	+0.004	[-0.004, +0.012]	1.000	–
heart	auc	SemanticLoss	+0.005	[-0.004, +0.014]	1.000	–
heart	ece	RandomForest	+0.021	[+0.006, +0.034]	0.044	✓
heart	ece	MLP	+0.003	[-0.010, +0.016]	1.000	–
heart	ece	SemanticLoss	-0.004	[-0.017, +0.009]	1.000	–
heart	logic_fid	RandomForest	+0.024	[-0.001, +0.050]	0.382	–
heart	logic_fid	MLP	+0.039	[+0.003, +0.080]	0.382	–
heart	logic_fid	SemanticLoss	+0.036	[+0.007, +0.070]	0.382	–
heart	rule_f1	RandomForest	+0.007	[-0.032, +0.046]	0.854	–
heart	rule_f1	MLP	+0.047	[+0.008, +0.093]	0.161	–
heart	rule_f1	SemanticLoss	+0.062	[+0.030, +0.094]	0.042	✓
pima	acc	RandomForest	-0.006	[-0.030, +0.016]	1.000	–
pima	acc	MLP	+0.006	[-0.014, +0.029]	1.000	–
pima	acc	SemanticLoss	-0.010	[-0.024, +0.005]	0.602	–
pima	acc_opt	RandomForest	+0.006	[-0.003, +0.016]	0.918	–
pima	acc_opt	MLP	-0.003	[-0.010, +0.004]	0.924	–
pima	acc_opt	SemanticLoss	+0.007	[-0.005, +0.020]	0.924	–
pima	f1	RandomForest	-0.008	[-0.022, +0.005]	0.905	–
pima	f1	MLP	-0.001	[-0.015, +0.013]	0.905	–
pima	f1	SemanticLoss	-0.010	[-0.020, +0.000]	0.396	–
pima	auc	RandomForest	+0.015	[+0.005, +0.025]	0.036	✓
pima	auc	MLP	+0.002	[-0.005, +0.008]	0.475	–
pima	auc	SemanticLoss	+0.005	[-0.001, +0.010]	0.228	–
pima	ece	RandomForest	+0.001	[-0.011, +0.011]	1.000	–
pima	ece	MLP	+0.001	[-0.007, +0.009]	1.000	–
pima	ece	SemanticLoss	-0.000	[-0.009, +0.009]	1.000	–
pima	logic_fid	RandomForest	+0.011	[-0.047, +0.068]	1.000	–
pima	logic_fid	MLP	+0.010	[-0.040, +0.064]	1.000	–
pima	logic_fid	SemanticLoss	-0.023	[-0.067, +0.019]	1.000	–

Table 12: Functional rule-behaviour probe on tic-tac-toe (5 seeds). Mean predicted win probability for valid winning lines, lines with one cell broken, and random boards. A logic-faithful model should separate the first two.

Board pattern	Mean prob.	Std
valid line	1.000	0.000
broken line	0.747	0.104
random board	0.476	0.126